Cloud Computing

ADSAI - High Performance Computing And Cloud

Course Introduction

Academic Year 2025/2026

DATA SCIENCE AND ARTIFICIAL INTELLIGENCE – University of Trieste

Course Context in ADSAI

Where Cloud Computing fits

P1: Foundations

- Computer Architecture
- Parallel Programming
- Numerical Methods

P2: Systems

- HPC Systems and programming
- Cloud Computing ←
- Data Management

P3: Applications

- Scientific Computing
- Machine Learning
- Domain Projects

Cloud and HPC are converging in AWS, Azure, GCP all offer HPC instances.
Understanding both paradigms is essential for modern scientific computing.
Our approach is Cloud in the framework of HPC.

Why Cloud for HPC Practitioners?

Traditional HPC

- Fixed capacity clusters
- Long queue times
- Specialized hardware
- Batch processing model

Cloud Computing

- Elastic, on-demand resources
- Instant provisioning
- Diverse instance types
- Service-oriented architecture

Why you need both:

- HPC clusters increasingly use cloud technologies (containers, orchestration)
- Cloud providers offer HPC-grade instances (InfiniBand, GPUs, bare metal)
- Hybrid workflows: develop in cloud, run production on HPC (or vice versa)
- Career flexibility: skills transfer between environments

Teaching Philosophy

"Don't be a user of pre-cooked tools that you consider as black-boxes"

❌ Not Just:

- Click buttons in AWS console
- Copy-paste Kubernetes YAML
- Follow tutorials blindly

✓ But Understand:

- WHY systems behave as they do
- HOW to debug distributed failures
- WHEN to choose which trade-off

Principles first, tools second. Understanding enables adaptation.

Learning Objectives

By the end of this course, you will be able to:

- Explain what is a Cloud environment
- Design and use Virtualized resources (local or cloud based)
- Design and deploy containerized applications (Docker, Compose)
- Deploy SLURM clusters using VM, containers and cloud services
- Deploy and manage applications on Kubernetes (Pods, Deployments)
- Provision cloud infrastructure programmatically (Terraform, AWS)
- Run HPC workloads in containers using Apptainer with MPI
- Evaluate cloud vs HPC trade-offs for scientific workloads

Course Structure: ~ 12 Lectures

Foundations
L1-L2
Distributed Systems
Cloud Models

→

Containers & Orch.
L3-L6
VMs, Docker
OpenStack

→

Cloud Infra
L7-L8
AWS, EC2
VPC, Terraform

→

HPC & Advanced
L9-L12
SLURM, Apptainer
Serverless, Integration

Exercise Progression:

Local (VMs, Docker, K8s) → Cloud (OS, AWS, Terraform) → HPC Integration (SLURM, Apptainer) → Exam

HOW IT WORKS

Lectures: we prepare slides, Lectures and Examples that you find on git

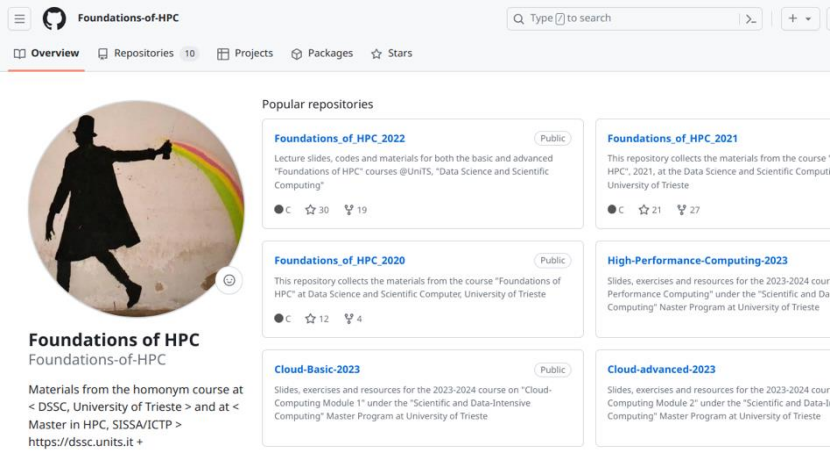
Ask questions, always: Understanding the fundamentals matters more than covering everything. If something isn't clear, stop me your question is likely shared by ten others.

Discussion: you ask to discuss much deeper some aspects or topics from previous/future lectures; you read other materials and bring them into the discussions.

Office hours: we are always ready for chatting, Q&A or alike upon appointment

HOW IT WORKS

Lectures: we prepare slides, Lectures and Examples that you find on git



`github.com/Foundations-of-HPC`

The repos are online and public; in the landing page you find the detailed list of the lectures.

Each lecture: ~45m it may include exercises

Lectures and Exercises

Complete lecture notes available: This year, each lecture comes with comprehensive notes on the course Git repository, not just slides. Notes include recommended books, websites, blogs, and videos for deeper exploration of each topic.

Practice exercises for self-assessment: Each lecture includes exercises designed to guide your learning. These are not required for the exam but help you gauge your understanding and identify gaps before the final assessment.

HOW IT WORKS: Hands-on Approach

Exercises

Sometimes 1-hour guided during lectures

CINECA OpenStack AWS Free Tier

Real cloud experience using AWS Free Tier. You'll set up your own account.

Tools

Docker, kubectl, AWS CLI, Apptainer.
Industry-standard tools.

⚠ AWS Account are NOT Required

I will show how to create a personal AWS account (free tier).
Always clean up resources after exercises to avoid charges!
But it is not necessary for the course

⚠ CINECA Account Required

Exame will require CINECA account to use Apptainer containers.

Assessment: the Final Exam

Eventually, this journey ends with an exam and an evaluation of your level of comprehension and acquired skills.

The evaluation consists in 2 steps

1) You complete a final assignment that will be given before the end of the lecture;
2) You attend the exam session
   A. every candidate will discuss individually its own assignment and the answers to the questions we ask about the assignment and the course
3) This year's **final exam covers HPC and Cloud together**. No more separate sessions: one comprehensive assessment that mirrors how these skills integrate in real-world scientific computing.

Exam Requirements:

- Individual work (collaboration allowed but submissions must be individual)
- Working code and infrastructure (must be demonstrable)
- Technical report documenting design decisions and results
- Brief presentation of your solution

Assessment: the Final Exam

Exam Requirements:

HPC will assign one or more exercises to be completed on CINECA's supercomputer. You must then replicate the same exercises using containers with Apptainer, and provide **an analytical and critical comparison** of results between bare-metal HPC and containerized execution.
Focus on performance differences, reproducibility, and trade-offs.

Prerequisites & Resources

Prerequisites

- Linux command line proficiency
- Basic networking (IP, ports, SSH)
- Programming experience (any language)
- Git version control basics
- HPC basics (SLURM, MPI concepts)

Resources

- Course repository (GitHub)
- Lecture notes (Markdown)
- Exercise materials with solutions
- AWS Setup Guide
- Reference book recommendations

Recommended Reference

Kleppmann, M. (2017). Designing Data-Intensive Applications. O'Reilly.
Excellent for distributed systems foundations - highly recommended!

Let's Begin!

Questions before we start?

Next: Lecture 1
Cloud Foundations & Distributed Systems